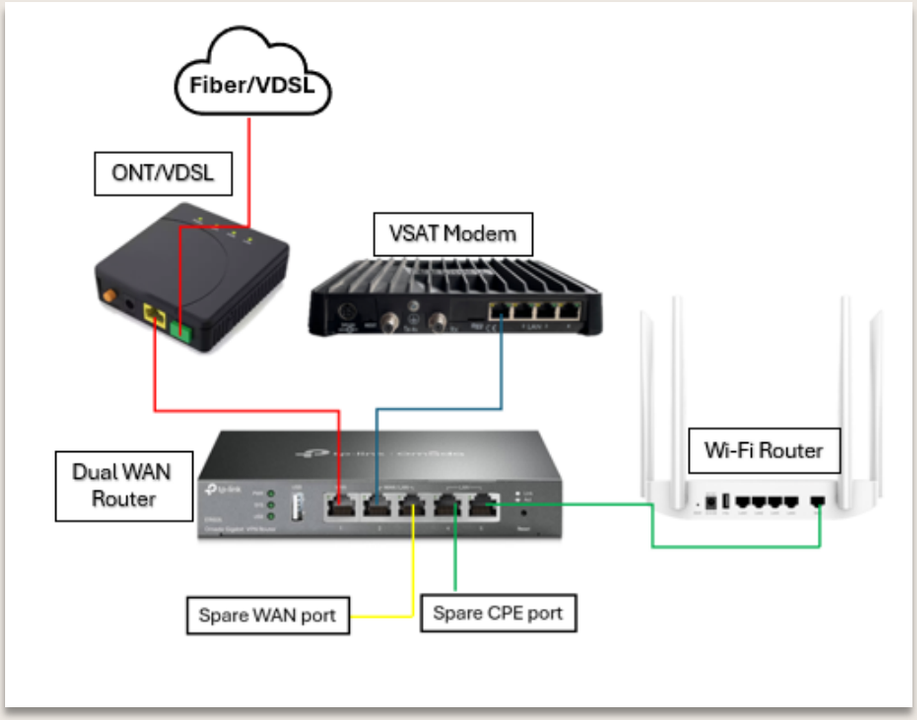
Troubleshooting Guide

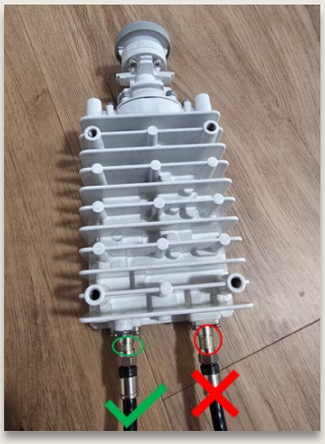
Fiber/VDSL + VSAT Product

Overview on the connection between ONT/VDSL modem, Dual WAN Router, VSAT Modem and Wi-Fi Router:



Choose the Broadband Issue You Are Experiencing:

1) Issue 1 – No Internet Connection

Step 1	Step 1: Check the RF Cable for Damage or Loose Connections
	
	• Ensure the RF cable has no kinks, cuts, or physical damage.
	• Make sure all cables are securely connected to the modem and transceiver:
	  ○ Screw in the RF cable fully by turning it clockwise.
	  ○ Gently shake the connection—it should be firm and not loose.
		
		 

Step 2	• Turn off the **VSAT modem**, **ONT/VDSL**, **Dual WAN Router**, and **Wi-Fi router** at the power source.
	• Wait at least **2 minutes** to ensure all devices fully reset.
	• Turn the power back on in the following order (if possible):
	  1. VSAT Modem / ONT / VDSL
	  2. Dual WAN Router
	  3. Wi-Fi Router	

<table>
<tr>
<td>Step 3</td>
<td>Check Connected Ports and LED Indicators

• Ensure all connected devices are powered on.
• Verify that all connected ports have a flashing green LED, indicating active data transmission.
• Inspect Ethernet cables for any damage and replace them if necessary.
• If the issue persists:
  ○ For a faulty Port 1 or 2, try connecting the Ethernet cable to Port 3 and contact Volt Broadband Support to activate it.
  ○ If Port 4 is faulty, switch to Port 5.</td>
<td>

<table>
<tr><th>Port</th><th>Description</th></tr>
<tr><td>1</td><td>Fiber/VDSL Connection</td></tr>
<tr><td>2</td><td>VSAT Connection</td></tr>
<tr><td>3</td><td>Spare WAN port</td></tr>
<tr><td>4</td><td>Wi-Fi Router connection</td></tr>
<tr><td>5</td><td>Spare Customer premise equipment port.</td></tr>
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<tr>
<td>Step 4</td>
<td>Ensure the Connection is to the WAN Port of the Wi-Fi Router

• Use an Ethernet (RJ45) cable to connect the Dual WAN router to the WAN/Internet port on your Wi-Fi router.
• This port is typically labeled “WAN” or “Internet” on the back of the router.
• For normal operation:</td>
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<ul>
<li>**Power** and **Wi-Fi** LEDs should be **solid**.</li>
<li>The **WAN** LED should be **flashing**, indicating active data transfer.</li>
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<tr>
<td>**Step 4**</td>
<td>Check your Modem LED status.</td>
<td>

LED Indicators and Descriptions

<table>
<tr><th></th><th>Lighting Warning</th><th>Description</th></tr>
<tr><td>1</td><td>Power LED</td><td>White continuous - when powered up.</td></tr>
<tr><td>2</td><td>RX indicator LED</td><td>White continuous - forward satellite signalling receiving.</td></tr>
<tr><td>3</td><td>TX indicator LED</td><td>White blinking / continuous - traffic transmitted via the satellite link.</td></tr>
<tr><td>4</td><td>Warning LED</td><td>Yellow continuous - when the terminal is not logged on to the satellite network.</td></tr>
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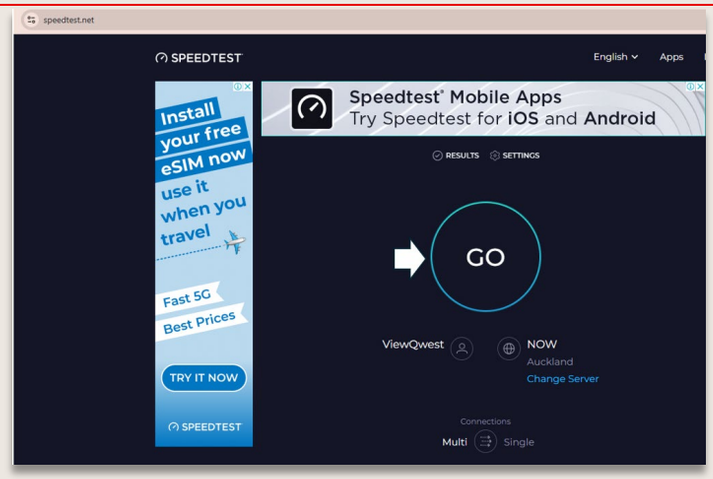
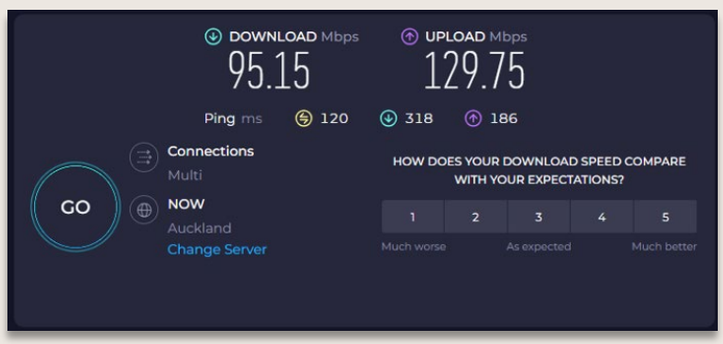
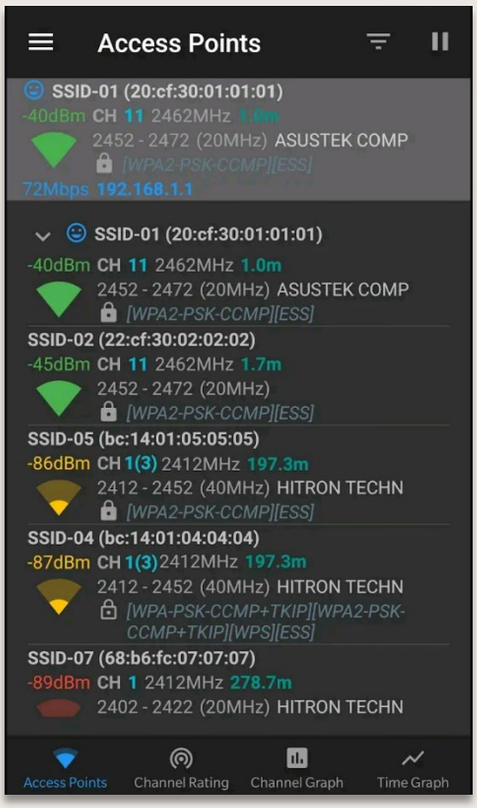
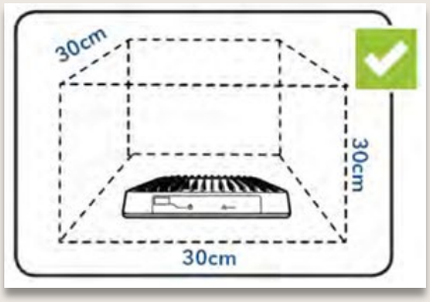
Issue 2 – Slow Internet Speed

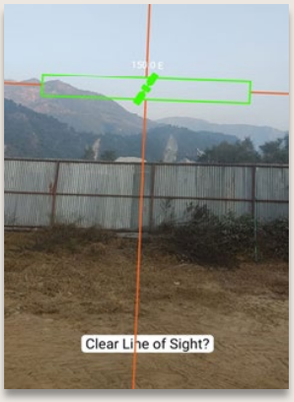
Step 1	**Perform a Speed Test** • Connect your laptop directly to **Port 4 or 5** of the **Dual WAN router** using an Ethernet cable, or use Wi-Fi (ensure your device is close to the router for the best signal). • Open a web browser and go to www.speedtest.net. • Click **"Go"** to begin the test. • Run the speed test **three times** to get a more accurate average result.	 

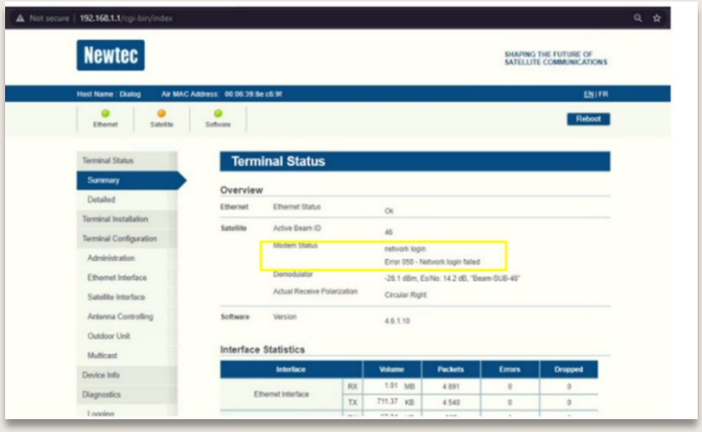
Step 2	**Improve Your Speed if Results Are Unsatisfactory**
	
	• **Disconnect unnecessary devices** from the network to free up bandwidth, then rerun the speed test.
	• **Optimize your Wi-Fi router's placement:**
	  ○ Place it in a **central, open location**, away from metal objects, appliances, or other electronics.
	  ○ Avoid placing it near **thick walls** or behind large furniture, which can block the signal.
	• **Check Wi-Fi signal strength:**
	  ○ Weak signals can reduce internet speed. Use tools like **Wi-Fi Analyzer** to check signal strength.
	  ○ If the signal is **below -70 dBm**, move your device closer to the router or reposition the router for better coverage.
		

Issue 3 – Frequent Disconnection

Step 1	Check for Overheating • Touch the equipment—if it feels hot, overheating may be the issue. • Ensure at least **30x30x30 cm of free space** around the equipment for proper heat dissipation.	

Issue 4 – Problems with the Satellite Modem

Step 1	Check for Rain Fade or Signal Loss
	
	• **Heavy rain** can weaken the satellite signal, causing slow speeds or disconnections. The connection should improve once the weather clears.
	• **Check the antenna's line of sight**:
	  ○ Ensure there are **no obstructions** (trees, buildings, etc.).
	  ○ Clear **snow or debris** from the antenna if necessary.
		
		 

Step 2	Access the Modem GUI for Further Diagnostics • **Connect your laptop directly to the modem** via Ethernet, or use Wi-Fi through the connected router. • Open a web browser and enter **192.168.1.1** to access the modem's **GUI (Graphical User Interface)**. • Navigate to the **Summary Page** and **share the status details** with our support team.	

For further assistance, contact **Volt Broadband Support**:
📧 **support@voltbroadband.co.nz**